50Days-coding-challenge

Day-19[SQL]

Farm Yield Optimization

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You are a data analyst at "Agri-Innovate," a company that provides data-driven solutions to farmers. Your mission is to help a client, a large-scale agricultural operation, analyze their farm data to improve crop yields and optimize resource allocation. The farmer wants to understand the relationship between different factors like planting, irrigation, and harvest to make better decisions for the next season.

The Database Schema

You have been given access to a database with the following four tables:

- farmers: Contains information about the farmers managing different plots.
    - farmer_id (INT, PRIMARY KEY): Unique ID for each farmer. ○ first_name (VARCHAR): Farmer's first name. ○ last_name (VARCHAR): Farmer's last name. ○ email (VARCHAR): Farmer's email address.
    - hire_date (DATE): Date the farmer was hired.
- plots: Contains details about each agricultural plot.
    - plot_id (INT, PRIMARY KEY): Unique ID for each plot. ○ plot_name (VARCHAR): Name of the plot (e.g., 'West Field', 'North Pasture'). ○ farmer_id (INT, FOREIGN KEY): ID of the farmer managing the plot. ○ crop_type (VARCHAR): Type of crop grown on the plot (e.g., 'Wheat', 'Corn', 'Soybeans').
    - soil_type (VARCHAR): Soil type of the plot (e.g., 'Loam', 'Clay', 'Sand').
- yields: Records crop yield data for each harvest.

- yield_id (INT, PRIMARY KEY): Unique ID for each yield record. ○ plot_id (INT, FOREIGN KEY): ID of the plot where the yield was recorded. ○ harvest_date (DATE): Date of the harvest. ○ yield_kg (DECIMAL): Yield in kilograms.
- weather_condition (VARCHAR): General weather condition during the growing season (e.g., 'Sunny', 'Rainy', 'Mild').

- irrigation_logs: Tracks irrigation events.
  - log_id (INT, PRIMARY KEY): Unique ID for each log entry. ○ plot_id (INT, FOREIGN KEY): ID of the plot that was irrigated. ○ irrigation_date (DATE): Date of the irrigation.
  - water_amount_liters (DECIMAL): Amount of water used in liters.

The Tasks

Write **SQL queries** to answer the following questions. Provide the queries and the resulting tables.

1. Productivity & Performance

- Identify the **top 3 most productive plots** based on average yield per harvest. Show the plot_name, crop_type, and average_yield_kg.

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No limit

Query | Query History

```sql
(17,17,'2025-05-04',1080),
(18,18,'2025-05-05',960),
(19,19,'2025-06-02',910),
(20,20,'2025-06-03',1450);

SELECT p.plot_name, p.crop_type, AVG(y.yield_kg) AS average_yield_kg
FROM plots p
JOIN yields y ON p.plot_id = y.plot_id
GROUP BY p.plot_name, p.crop_type
ORDER BY average_yield_kg DESC
LIMIT 3;
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 | Page No: 1 of 1

	plot_name character varying (100)	crop_type character varying (50)	average_yield_kg numeric
1	Silver Field	Rice	650.0000000000000000
2	Delta Farm	Rice	640.0000000000000000
3	Rain Field	Rice	630.0000000000000000

Total rows: 3 | Query complete 00:00:00.624 | CRLF | Ln 124, Col 1

- Calculate the **total water consumption** for each plot and rank them from highest to lowest. Show plot_name and total_water_liters.

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No limit

Query | Query History

```sql
FROM plots p
JOIN yields y ON p.plot_id = y.plot_id
GROUP BY p.plot_name, p.crop_type
ORDER BY average_yield_kg DESC
LIMIT 3;

SELECT p.plot_name, SUM(i.water_amount_liters) AS total_water_liters
FROM plots p
JOIN irrigation_logs i ON p.plot_id = i.plot_id
GROUP BY p.plot_name
ORDER BY total_water_liters DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 20 | Page No: 1 of 1

	plot_name character varying (100)	total_water_liters numeric
1	Silver Field	1450.00
2	Delta Farm	1400.00
3	Rain Field	1350.00
4	River Side	1300.00
5	Sunset Plot	1250.00
6	North Field	1200.00

Total rows: 20 | Query complete 00:00:00.360 | CRLF | Ln 131, Col 1

2. Yield & Environmental Analysis

- Determine the **average yield for each crop type** under different weather conditions. The output should have crop_type, weather_condition, and average_yield_kg.

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No limit

Query | Query History

```
ORDER BY total_water_liters DESC;

SELECT p.crop_type, y.weather_condition, AVG(y.yield_kg) AS average_yield_kg
FROM plots p
JOIN yields y ON p.plot_id = y.plot_id
GROUP BY p.crop_type, y.weather_condition;
```

Data Output | Messages | Notifications

Showing rows: 1 to 8 | Page No: 1 of 1

	crop_type character varying (50)	weather_condition character varying (50)	average_yield_kg numeric
1	Rice	Rainy	627.5000000000000000
2	Soybeans	Rainy	480.0000000000000000
3	Corn	Mild	440.0000000000000000
4	Soybeans	Mild	470.0000000000000000
5	Wheat	Sunny	532.5000000000000000
6	Corn	Sunny	436.6666666666666667
7	Wheat	Rainy	510.0000000000000000
8	Rice	Sunny	620.0000000000000000

Total rows: 8 | Query complete 00:00:00.289 | CRLF | Ln 137, Col 1

- Find the **highest-yielding plot for each soil type**. Show the soil_type, plot_name, and highest_yield_kg.

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The session is idle and there is no current transaction.

Query | Query History

```sql
GROUP BY p.crop_type, y.weather_condition;

SELECT p.soil_type, p.plot_name, MAX(y.yield_kg) AS highest_yield_kg
FROM plots p
JOIN yields y ON p.plot_id = y.plot_id
GROUP BY p.soil_type, p.plot_name
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 | Page No: 1 of 1

	soil_type character varying (50)	plot_name character varying (100)	highest_yield_kg numeric
1	Loam	Rain Field	630.00
2	Clay	Silver Field	650.00
3	Sand	Fresh Land	480.00

Total rows: 3 | Query complete 00:00:00.207 | CRLF | Ln 142, Col 1

3. Farmer & Resource Management

- Identify the **farmer who manages the plots with the lowest average water consumption**. Show the first_name, last_name, and their plots' average_water_liters_per_plot.

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No limit

Query | Query History

```sql
    WHERE p2.soil_type = p.soil_type
);

SELECT f.first_name, f.last_name, AVG(i.water_amount_liters) AS avg_water_liters
FROM farmers f
JOIN plots p ON f.farmer_id = p.farmer_id
JOIN irrigation_logs i ON p.plot_id = i.plot_id
GROUP BY f.farmer_id, f.first_name, f.last_name
ORDER BY avg_water_liters ASC
LIMIT 1;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page No: 1 of 1

	first_name character varying (50)	last_name character varying (50)	avg_water_liters numeric
1	Lakshmi	Devi	850.0000000000000000

Total rows: 1 | Query complete 00:00:00.202 | CRLF | Ln 153, Col 1

- Calculate the **number of harvests per month** for the last 12 months. Show the month and the number of harvests.

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No limit

Query | Query History

```
GROUP BY f.farmer_id, f.first_name, f.last_name
ORDER BY avg_water_liters ASC
LIMIT 1;

SELECT TO_CHAR(harvest_date, 'YYYY-MM') AS month,
       COUNT(*) AS number_of_harvests
FROM yields
WHERE harvest_date >= CURRENT_DATE - INTERVAL '12 months'
GROUP BY month
ORDER BY month;
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 | Page No: 1 of 1

	month text	number_of_harvests bigint
1	2025-04	2
2	2025-05	4
3	2025-06	2

Total rows: 3 | Query complete 00:00:00.141 | CRLF | Ln 161, Col 1

4. Advanced Analysis (Bonus)

- Find plots that have a **below-average yield** for their specific crop type but an **aboveaverage water consumption** compared to all other plots with the same crop. List the plot_name, crop_type, yield_kg, and water_amount_liters.

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No limit

Query | Query History

```
FROM yields
WHERE harvest_date >= CURRENT_DATE - INTERVAL '12 months'
GROUP BY month
ORDER BY month;

SELECT p.plot_name, p.crop_type, y.yield_kg, i.water_amount_liters
FROM plots p
JOIN yields y ON p.plot_id = y.plot_id
JOIN irrigation_logs i ON p.plot_id = i.plot_id
WHERE y.yield_kg < (
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	plot_name character varying (100)	crop_type character varying (50)	yield_kg numeric (10,2)	water_amount_liters numeric (10,2)
1	South Field	Wheat	520.00	1100.00
2	Valley Plot	Soybeans	470.00	980.00
3	Hot Plot	Corn	420.00	890.00

Total rows: 3 | Query complete 00:00:00.234 | CRLF | Ln 168, Col 1